ARDUINO BABY MONITOR

Smart Nursery Solution

SIMPLE • RELIABLE • AFFORDABLE

KEY FEATURES & CAPABILITIES

Environmental Monitoring

- Real-time temperature tracking (DHT11)
- Humidity monitoring for nursery health
- Ensures optimal sleeping conditions

Smart Cry Detection

- High-sensitivity sound sensor (KY-038)
- 3-reading confirmation logic
- Suppresses false alarms from room noise

Instant Alerts

- Audible buzzer for immediate attention
- Visual Red LED notification
- Triggers when safety thresholds are met

Clear Status Display

- 16x2 I2C LCD for live data output
- Human-readable status messages
- "Room OK" vs "Baby Crying" alerts

ESSENTIAL HARDWARE COMPONENTS

Category	Component	Purpose
Microcontroller	Arduino Uno / Nano	System Brain
Sensors	DHT11	Temp & Humidity
Sensors	KY-038	Cry Detection
Display	16x2 I2C LCD	Status Output
Alerts	Buzzer & Red LED	Audible/Visual

Technical Note: Sensors

The KY-038 module features an onboard sensitivity potentiometer to calibrate sound detection levels.

The DHT11 provides digital readings for both environmental factors via a single-wire interface.

CIRCUIT DESIGN & WIRING LOGIC

Component	Sensor Pin	Arduino Pin
DHT11	DATA	D2 (+10kΩ Pull-up)
KY-038	D0 (Digital)	D3
LCD I2C	SDA / SCL	A4 / A5
Buzzer	Positive (+)	D8
Red LED	Anode (+)	D9 (via 220Ω)

POWER DISTRIBUTION

VCC (5V Rail)	DHT11, KY-038, LCD
GND (Common)	All Components

COMMUNICATION

I2C Protocol	LCD Display
Digital Input	Sound & Temp

SOFTWARE ENVIRONMENT & LIBRARIES

1. IDE CONFIGURATION

01 Download and install **Arduino IDE 2.x**

02 Select Board: **Tools > Board > Arduino Uno**

03 Configure Port: **Tools > Port > [Your Port]**

04 Set Serial Monitor to **9600 Baud**

```
// Serial Initialization
Serial.begin(9600);
```

2. REQUIRED LIBRARIES

Library Name	Author	Purpose
DHT sensor library	Adafruit	Read Temp & Humidity
Adafruit Unified Sensor	Adafruit	DHT Dependency
LiquidCrystal_I2C	Frank de Brabander	Drive 16x2 I2C LCD
Wire.h	Built-in	I2C Communication

** Install via Sketch > Include Library > Manage Libraries*

CALIBRATION & INITIAL SETUP

I2C DISCOVERY

01

Identify the LCD's hardware address using an I2C scanner sketch.

Common addresses:

- **0x27** (Blue LCDs)
- **0x3F** (Green LCDs)

```
Wire.beginTransmission(addr);
if (Wire.endTransmission() == 0)
```

SOUND SENSITIVITY

02

Adjust the **KY-038** potentiometer to filter background noise.

Clockwise: Less sensitive (loud sounds only).

Counter-Clockwise: More sensitive (quieter sounds).

Target: Trigger on clap at 50cm.

LCD CONTRAST

03

Fine-tune the visual output for maximum readability.

Locate the blue potentiometer on the I2C backpack.

Adjust until text is sharp and background blocks disappear.

THE MONITORING LOGIC LOOP

1 Sensor Polling

Read DHT11 (Temp/Hum) and KY-038 (Sound) every 2 seconds.

2 Threshold Comparison

Compare readings against pre-defined safety ranges.

3 Cry Confirmation

Increment counter if sound is detected; decrement if quiet.

4 Output Update

Refresh LCD status and trigger Buzzer/LED if needed.

Smart Cry Detection

To prevent false alarms from brief noises, the system uses a **3-reading confirmation window**.

The baby must be crying consistently for **6 seconds** (3 polls × 2s) before the alert fires.

```
#define CRY_CONFIRM 3
#define READ_INTERVAL 2000
```

SAFE NURSERY THRESHOLDS

Recommended Ranges

Parameter	Safe Range
Temperature	18°C – 22°C
Humidity	40% – 60%
Sound Level	Ambient / Quiet

Based on NHS and AAP recommendations for infant sleep safety and mold prevention.

CONFIGURATION: baby_monitor.ino

```
// Environmental Thresholds
#define TEMP_HIGH     22.0
#define TEMP_LOW      18.0
#define HUM_HIGH      60
#define HUM_LOW       40

// Logic Settings
#define CRY_CONFIRM   3
#define READ_INTERVAL 2000
```

Thresholds are defined as global constants for easy end-user customization.

** Disclaimer: This project is a prototype. Always follow professional paediatric guidance for nursery monitoring.*

REAL-TIME FEEDBACK & STATUS MESSAGES

LCD Display Logic

Row 1 Message	System State
Room OK :)	All readings within safe range
!! BABY CRYING !!	Sound detected (Alarm Active)
ROOM TOO HOT!	Temp > 22°C
TOO HUMID	Humidity > 60%
Sensor error!	DHT11 communication failure

SERIAL MONITOR @ 9600 BAUD

```
=== Arduino Baby Monitor ===
Initialising sensors...
Ready. Monitoring started.

Temp: 20.3 C | Hum: 52 % | Sound: quiet
Temp: 20.4 C | Hum: 51 % | Sound: quiet
Temp: 20.4 C | Hum: 52 % | Sound: CRYING
Temp: 20.5 C | Hum: 52 % | Sound: CRYING
```

* Serial logging provides detailed telemetry for remote debugging and system verification.

TROUBLESHOOTING & FUTURE UPGRADES

Common Issues

Problem	Likely Cause / Fix
LCD Blank	Wrong I2C address or low contrast.
DHT11 'nan'	Missing 10kΩ pull-up resistor.
False Alerts	Sound sensitivity too high.
No Alerts	Sensitivity too low or wiring error.

FUTURE POTENTIAL

Wi-Fi Integration

Upgrade to ESP32 for mobile notifications via Telegram or Blynk.

Data Logging

Add an SD card module to track nursery trends over time.

Night Light

Integrate a PWM-controlled LED strip for a gentle nursery glow.

Web Dashboard

Serve a local HTML page with live charts and historical data.